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Introduction

This report aims to explore player performance and market value in professional football through the application of logistic regression and clustering techniques. Using detailed match statistics for midfielders, we analyzed players' contributions on the field and linked them to changes in their market value over consecutive seasons.

We first started the analysis by carefully cleaning and converting the data to ensure data consistency and dataset reliability. We thereafter applied Principal Component Analysis (PCA) and k-means clustering to identify distinctive profiles of midfielders that reflect playing style and patterns of performance.

Based on these results, we constructed a logistic regression model to predict whether the market value of any given player would increase or decrease, based on both standard performance metrics as well as on the player's cluster classification. This approach allows us to identify the most important factors driving changes in market value and to observe the types of players behind these variations.

Overall, this research provides a complete framework for football performance and market value analysis, combining high-level statistical modeling with data-driven player profiling. The final goal is to offer conclusions that potentially can inform scouting, development, and strategic decision-making in professional football.

2 Exploratory Analysis

2.1 Description of the Dataset

The dataset utilized in this work consists of match statistics of Premier League midfielders.

It helps to combine offense and defense data in a bid to present an understandable sense of player performance and how that can impact on future value.

To ensure comparability across players with different playing time, most raw statistics were normalized per minute of play. After preprocessing, the following variables were selected for the analysis:

- **Cmp%**: Pass completion percentage.
- **KP_min**: Key passes per minute.
- **xA_min**: Expected assists per minute.
- **PPA_min**: Passes into the penalty area per minute.
- **Sh/90**: Total shots per 90 minutes.
- **Dist_pass**: Average passing distance.
- **PrgP_pass**: Progressive passes per match.
- **Won%**: Percentage of defensive duels won.
- **Pass_min**: Total passes per minute.
- **Tkl_min**: Tackles per minute.
- **Blocks_min**: Defensive blocks per minute.
- **Int_min**: Interceptions per minute.
- **Clr_min**: Clearances per minute.
- **Recov_min**: Recoveries per minute.
- **Fls_min**: Fouls committed per minute.
- **xG_min**: Non-penalty expected goals per minute.
- **CPA_min**: Completed passes into the attacking area per minute.
- **Def_min**: Defensive third actions per minute.
- **Att_min**: Attacking third actions per minute.
- **Pen_min**: Entries into the penalty area per minute.
- **PrgDist_min**: Progressive distance carried with the ball per minute.

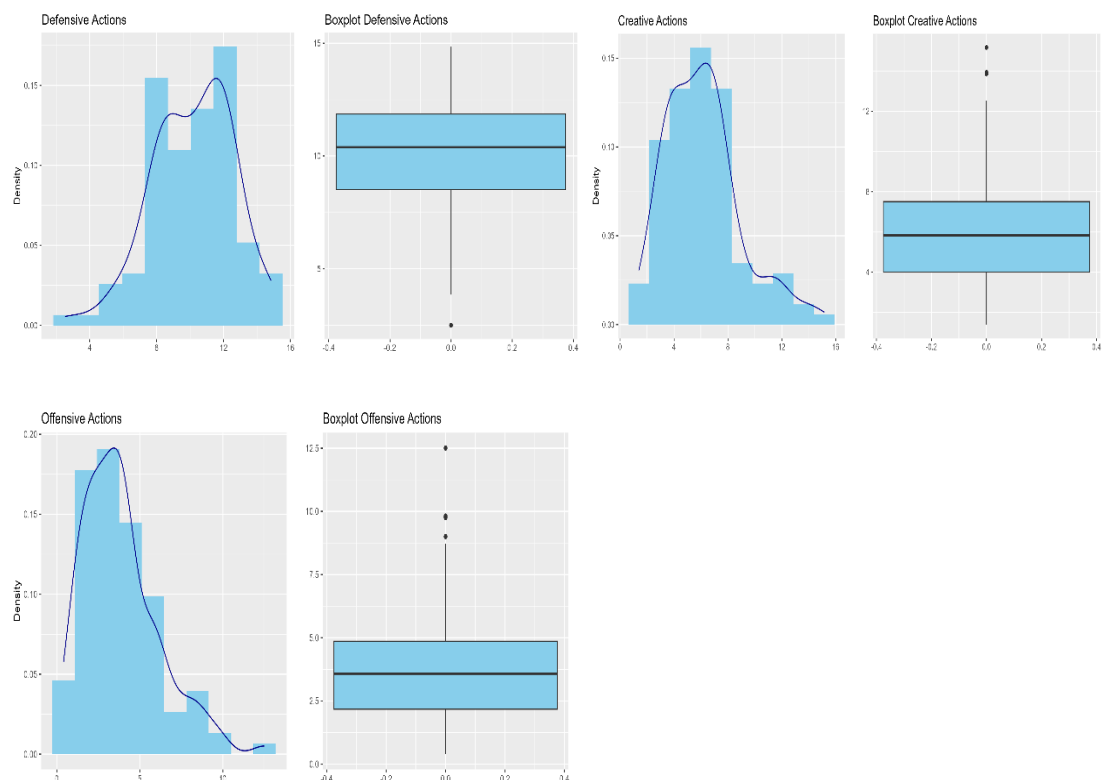
This set of variables provides an even-sided snapshot of players' technical, creative, and defensive performance. Through combining passing efficiency, chance creation, defensive contribution, and progression statistics, the data enables a multi-dimensional evaluation of player roles and effectiveness.

2.2 Feature Distributions

The distributions of the main performance indicators per 90 minutes help us understand player tendencies and detect outliers.

- **Defensive Actions (per 90):** This category comprises tackles, interceptions, blocks, clearances, and recoveries. The histogram is left skewed, reflecting that the majority of midfielders have a moderate to high amount of defensive actions, with almost none having extremely low values.
- **Creative Actions (per 90):** Records passing and creativity-creating ability (key passes, expected assists, progressive passes). The distribution is skewed to the right, meaning most midfielders have a medium level of creativity, but few have very high creative output.
- **Offensive Actions (per 90):** This measure, a combination of shots, expected goals, and goals, is an indication of the attacking contribution of midfielders. The distribution is very right skewed, highlighting the fact that offensive contribution is normally low for most midfielders with only a minor subgroup having very high attacking contribution.

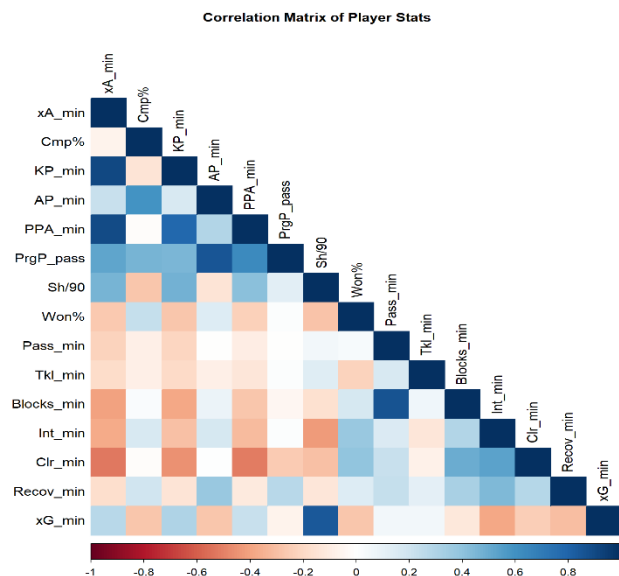
The boxplots also support these findings with higher spread and more extreme outliers for offensive contribution than defensive or creative ones. This shows that defensive and creative contributions are more spread out among midfielders, but offensive involvement is in a smaller subset of players.



2.3 Correlation Analysis

To better understand the relationships among the selected performance indicators, a correlation matrix was computed and visualized. The analysis reveals several noteworthy patterns:

- **Creative passing metrics** such as *xA*, *Key Passes* and *Passes into Penalty Area* show very strong positive correlations ($\rho \approx 0.78\text{--}0.90$), suggesting that players who create expected assists also tend to generate key passes and progress the ball into advanced areas.
- **Shooting activity (Shots/90) and xG** display a high positive correlation ($\rho = 0.85$), reflecting that players who attempt more shots generally also accumulate higher expected goals.
- **Defensive actions** (Tackles, Interceptions, Clearances, Blocks, Recoveries) are positively correlated with each other but negatively correlated with creative/offensive metrics, indicating a specialization trade-off: midfielders are either more defensively oriented or more offensively/creatively inclined.
- **Passing accuracy (Cmp%)** correlates positively with progressive and advanced passing (*AP*, *PrgP_pass*), underlining that technically secure players are also more effective in advancing play.



Overall, the correlation structure highlights two main axes of specialization: creative contribution, where metrics cluster strongly together, defensive contribution, where actions also cluster but remain inversely related to the first group, and offensive , where *Shots/90* and *xG* are very strongly correlated ($\rho \approx 0.85$), representing players with a direct attacking output.

3.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimension-reduction method that enables us to condense a large number of correlated variables into a few uncorrelated components while maintaining most of the original data information. In this research, PCA was used to some of the most important performance indicators for midfielders, such as passing, attacking, and defensive data.

The analysis produced multiple principal components (PCs), each representing a linear combination of the original variables. The first few components capture the majority of the variance in the data:

Component	Standard Deviation	Proportion of Variance	Cumulative Variance
PC1	2.684	32.7%	32.7%
PC2	2.031	18.8%	51.5%
PC3	1.493	10.1%	61.6%
PC4	1.295	7.6%	69.2%
PC5	1.084	5.3%	74.6%

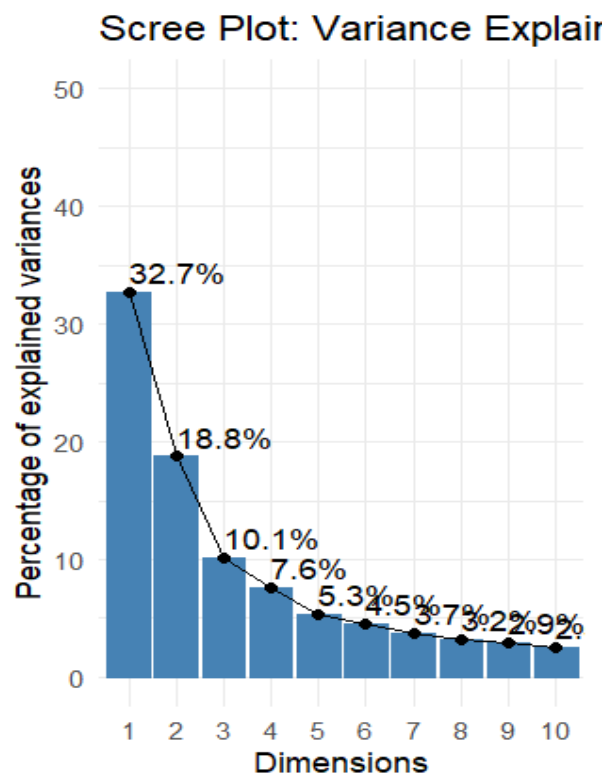
Based on these results, the first three components were retained for further analysis. They collectively explain approximately 62% of the total variance, which provides a good balance between dimensionality reduction and information retention.

Interpretation of Components:

- **PC1 – Attacking and Playmaking:** This component is strongly influenced by key passes, progressive passes, expected goals, and shots per 90 minutes. Players with high scores on PC1 are typically creative midfielders who contribute significantly to offensive actions.
- **PC2 – Passing Efficiency and Control:** PC2 loads heavily on passing accuracy, short passes, and progressive passing. Midfielders with high PC2 scores are often stable playmakers who control possession and distribute the ball effectively.
- **PC3 – Defensive Contribution:** PC3 places greater emphasis on tackling, interceptions, and clearances. PC3 goal-scoring players are defensive midfielders who are experts at regaining possession of the ball and protecting the back line.

These three principal components were then used as input for clustering, allowing us to identify groups of players with similar performance profiles while reducing noise and redundancy in the original dataset.

The choice of three components was also supported by the scree plot, which shows the eigenvalues of all components. A clear “elbow” appears at the third component, indicating that adding further components would yield diminishing returns in terms of explained variance.

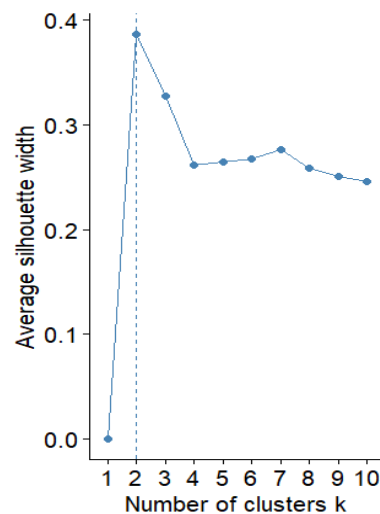


Scree plot of principal components showing variance explained by each component.

3.2 K-Means Clustering

Following the PCA, we applied **K-Means Clustering** to group midfielders based on the first three principal components (PC1, PC2, PC3), which together explain around 62% of the variance in the dataset.

The optimal number of clusters was explored using the elbow and silhouette methods. While the silhouette method suggested 2 clusters, the decision was made to use 3 clusters, as this solution better reflects the tactical roles observed in midfielders (defensive, attacking, and balanced).

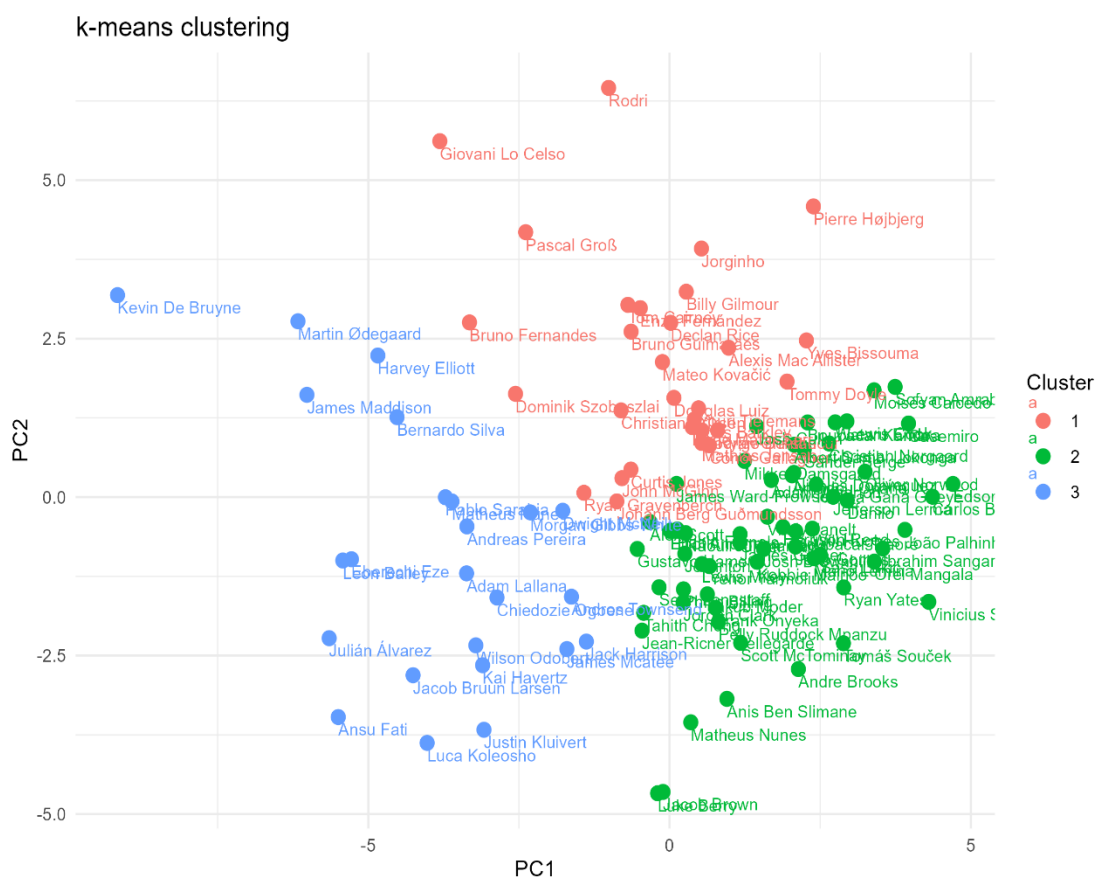


The players were divided into three groups:

- **Cluster 1 (29 players):** Balanced midfielders with contributions both in defense and attack. This cluster can be interpreted as *box-to-box midfielders* (e.g., Rodri, Declan Rice).
- **Cluster 2 (60 players):** Midfielders with more defensive involvement (e.g., interceptions, tackles, recoveries). This cluster corresponds to *defensive midfielders* (e.g., Casemiro).
- **Cluster 3 (24 players):** Attacking and creative midfielders (e.g., high xG, xA, shot creation). This cluster includes attacking/creative midfielders (e.g. Kevin De Bruyne).

Cluster	Mean PC1	Mean PC2	Mean PC3	Count
1	-0.25	2.20	-0.11	29
2	1.72	-0.70	-0.025	60
3	-4.00	-0.92	0.2	24

The k-means clustering plot shows three distinct groups of midfielders.



3.2 Logistic Regression for Predicting Player Value Changes

To calculate the probability of the market value of a midfielder to increase in the 2023/24 season, the performance data of the two preceding seasons (2021/22 and 2022/23) have been used to estimate a logistic regression model.

The target variable, called **Target**, is binary:

- Target = 1 if the player's market value increased from 2021/22 to 2022/23
- Target = 0 if the player's market value stayed the same or decreased

By modeling this binary outcome, the logistic regression estimates the probability that a player's value will rise in the following season based on recent performance trends.

The independent variables (predictors) were the differences in main performance indicators per 90 minutes between the two seasons, including:

- Offensive contributions: Key Passes (KP), Expected Goals (xG), Minutes Played (Min)
- Defensive contributions: Clearances (Clr), Won % (Won)
- Age at the end of the 2022/23 season (Age_22)

That is, the model connects performance statistic variations with the chance of a value increase, logging which performance measures matter most to valuation by the market.

The logistic regression model can be written as:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 KP + \beta_2 Won + \beta_3 Clr + \beta_4 xG + \beta_5 Min + \beta_6 Age_{\{22\}}$$

where p is the probability that a player's value increased.

Model Results

	Estimate	Std. Error	z value	p-value
(Intercept)	7.894	5.510	1.433	0.152
KP	176.4	126.7	1.392	0.164
Won	0.113	0.056	1.986	0.047 *
Clr	-160.8	93.4	-1.721	0.085 .
xG	1837.0	784.2	2.343	0.019 *

*Significance codes: **p < 0.01**, $p < 0.05$, . $p < 0.1$

The model shows a good fit with an AIC of 39.142, residual deviance of 25.142 on 39 degrees of freedom, and null deviance of 60.603 on 45 degrees of freedom.

Model Predictions on Training Data

The predicted probabilities for all players in the training set were calculated and classification was performed using a cut point of 0.5.

Predicted \ Actual	0	1
0	27	3
1	2	14

Logistic Regression Predictions for 2023/24 Season

The logistic regression model was trained on player data from previous seasons (21/22 and 22/23) to predict whether a player's market value would increase in the 2023/24 season.

Using the model, we estimated predicted probabilities and corresponding classes (1 = increase, 0 = decrease) for 25 selected midfielders.

The following table shows the actual outcomes and the model predictions for each player for the 2023/24 season:

Player	Actual	Predicted	Probability
Rodrigo Bentancur	0	0	0.015
Yves Bissouma	1	1	0.967
Moisés Caicedo	0	0	0.005
Kevin De Bruyne	0	0	0.000
Harvey Elliott	1	1	0.994
Christian Eriksen	0	0	0.000
Bruno Fernandes	0	0	0.068
Pablo Fornals	0	0	0.000
Morgan Gibbs-White	0	1	0.930
Billy Gilmour	1	1	0.998
Pascal Groß	0	0	0.348
Bruno Guimarães	1	1	0.755
Mathias Jensen	1	0	0.001
Joelinton	0	0	0.001
Mateo Kovačić	0	0	0.034
Alexis Mac Allister	1	0	0.002
James Maddison	1	0	0.152
Scott McTominay	1	1	0.982
Declan Rice	1	0	0.376
Rodri	0	0	0.290
Bernardo Silva	0	1	0.702
Oliver Skipp	0	0	0.201
Youri Tielemans	0	0	0.010
James Ward-Prowse	0	0	0.035
Martin Ødegaard	1	0	0.042

This table shows which players are expected to see a rise in value (class 1) and which are expected to decrease (class 0). The fitted probability is the measure of how confident the model is with each prediction.

Model Performance

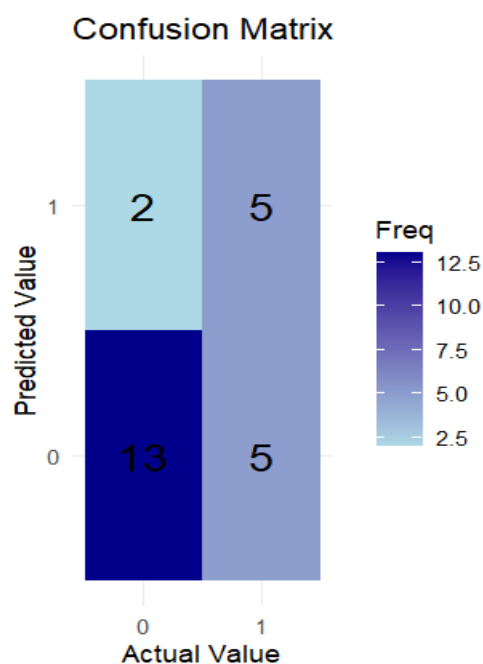
The model was able to predict 18 out of 25 players accurately, providing an accuracy of approximately 72%.

This suggests a moderate level of predictive performance, although there were a few misclassifications — specifically in cases where the player performance changes were small or unpredictable.

These discrepancies suggest that market value is partly dependent not just on statistical performance but also qualitative and external factors (e.g., team context, injuries, media attention, or contract status).

Confusion Matrix

The confusion matrix below provides a clearer picture of the model's performance. It shows how many increases/decreases were correctly or incorrectly predicted:



Overall, the logistic regression model provides a useful but limited prediction framework for player value changes.

It captures general trends, but it misses some cases where non-performance-related factors play a role.

Future improvement could include the addition of team level attributes, recent injury tracking, or transfer history, which could make the predictions more realistic and in accordance with actual market trends.